

Project Baseball: Monte Carlo Simulation

Rey Wan, Erwin Vasquez, Nathanael Choi, Maxence Bolton



THE PROBLEM



- 🏈 **Motivating question:** How do we develop a robust model to simulate a series between two MLB teams?
- 🏈 **Goal:** Simulate a 3-game, 9-innings per game (barring extra innings in case of a tie) series between the **Baltimore Orioles** and **Chicago Cubs**
 - **NOT best of 3:** all 3 games are played no matter what, even if a team is up 2-0
 - Simulate MLB regular season series, where teams always play all scheduled games against each other
- 🏈 Focused on team batting statistics only from 2025 regular season games
 - Plate appearances, at bats, walks, singles, doubles, triples, HRs, etc.

Data



<u>Orioles</u>	<u>Cubs</u>
Plate Appearances: 6020	Plate Appearances: 6162
AB: 5416	AB: 5495
Walks (BB): 484	Walks (BB): 554
Singles (PA - 2B - 3B - HR): 812	Singles (PA - 2B - 3B - HR): 852
Doubles (2B): 251	Doubles (2B): 267
Triples (3B): 19	Triples: 29
Home Runs: 191	Home Runs: 223
Hit by pitch (HBP): 75	Hit by pitch (HBP): 46
Strikeouts (SO): 1457	Strikeouts (SO): 1277
Hits (H): 1273	Hits (H): 1371
Stolen Bases (SB): 121	Stolen Bases (SB): 161
Grounded into Double Plays: 102	Grounded into Double Plays: 72
Caught Stealing (CS): 37	Caught Stealing (CS): 35
Outs (PA- H - SO - BB - HBP): 2731	Outs (PA- H - SO - BB - HBP): 2914

COUNTS TO PROBABILITIES



Used Markov Analysis with each state determined by base runner positions and outs at the beginning of **each plate appearance**

As a result, needed to convert our batting statistics into **per plate appearance probabilities** to be suitable for modeling

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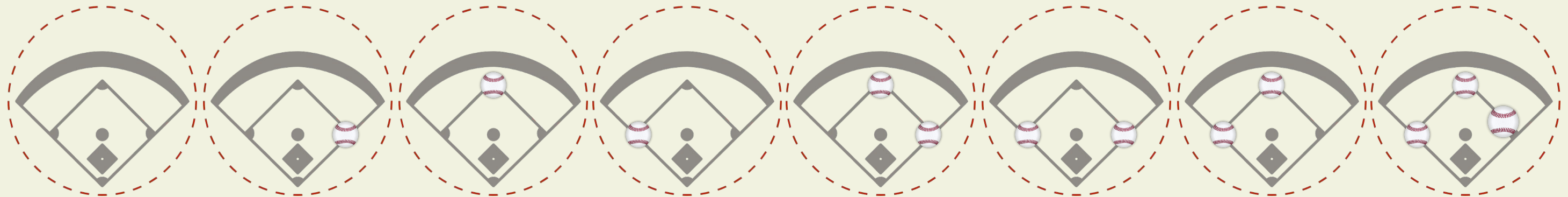
<u>Orioles</u>	<u>Cubs</u>
Plate Appearances: 6020	Plate Appearances: 6162
AB: 5416	AB: 5495
Walks (BB): $484/6020 = .0804$	Walks (BB): $554/6162 = .09$
Singles: $812/6020 = .1349$	Singles: $852/6162 = .138$
Doubles (2B): $251/6020 = .0417$	Doubles (2B): $267/6162 = .043$
Triples (3B): $19/6020 = .00316$	Triples: $29/6162 = .005$
Home Runs: $191/6020 = .0317$	Home Runs: $223/6162 = .0362$
Hit by pitch (HBP): $75/6020 = .0125$	Hit by pitch (HBP): $46/6162 = .007$
Strikeouts (SO): $1457/6020 = .2420$	Strikeouts (SO): $1277/6162 = .207$
Hits (H): $1273/6020 = .201$	Hits (H): $1371/6162 = .222$
Stolen Bases (SB): $121/6020 = .021$	Stolen Bases (SB): $161/6162 = .026$
Grounded into Double Plays: $102/6020 = .01694$	Grounded into Double Plays: $72/6162 = .01168$
Caught Stealing (CS): $37/6020 = .00615$	Caught Stealing (CS): $35/6162 = .005679$
Outs: $4188/6020 = .696$	Outs: $4191/6162 = .680$
Outs - GIDP: $4086/6020 = .679$	Outs - GIDP: $4119/6162 = .668$

MARKOV CHAIN

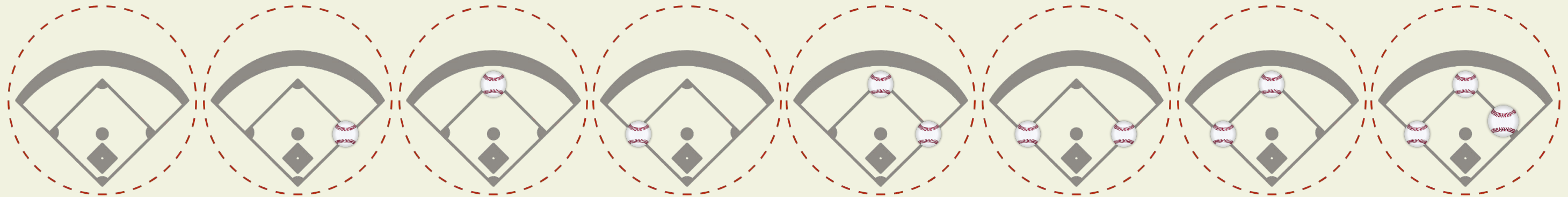


25 STATES

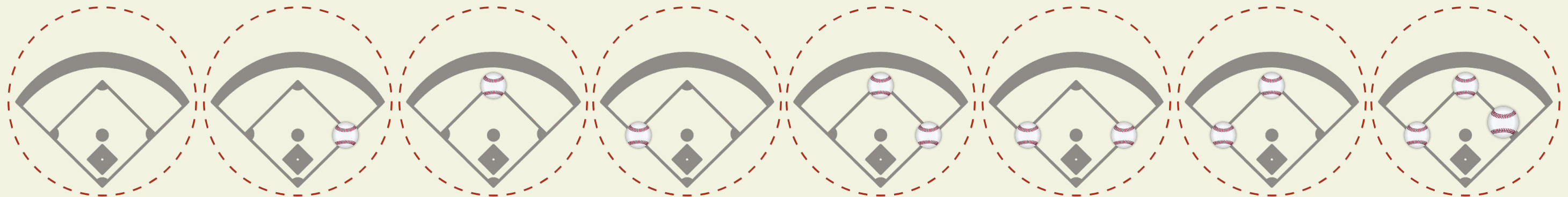
0 OUTS:



1 OUT:



2 OUTS:



3 OUTS:



MODELING ASSUMPTIONS



- ⚾ **Conservative baserunning:** runners advance as many bases as the batter does
 - **No Tagging Up:** runners can't advance on an out
- ⚾ **No Errors:** no misplays that let batters/runners reach or take extra bases
- ⚾ **No Triple Plays**
- ⚾ **Assume “force out”:** the out will always be for the furthest base runner
- ⚾ **Double Plays:** must be force outs; will always occur at the furthest base runners, EXCEPT for 0 outs, (123)
- ⚾ **Stealing:** a baserunner will only steal second base, always successfully
- ⚾ **Memoryless:** no previous games affect future games
- ⚾ **Context neutral:** probability of a given outcome is constant regardless of baserunners or outs

DOUBLE PLAYS & STOLEN BASES



Double Plays (DP)

- 🏈 Treat DP as a subcategory of OUT $DP \subset OUT \rightarrow \text{set } pOUT = pOUT - pDP$
- 🏈 Add DP transitions 0 outs, 1- - to 2 outs, - - - and 1 out, 1- - to 3 outs (end)
- 🏈 Bases loaded rule: $pDP=0$ for 0 outs, 123; allow only 1 out, 123 $\rightarrow$ 3 outs

Stolen Bases (SB)

- 🏈 Stolen bases are only considered when second base is free
- 🏈 Included into the probability in moving to the state with a runner on second after there is a runner on first.
- 🏈 When bases are full (123) $pSB=0$

P MATRIX: ORIOLES

[illegible]

P MATRIX: CUBS



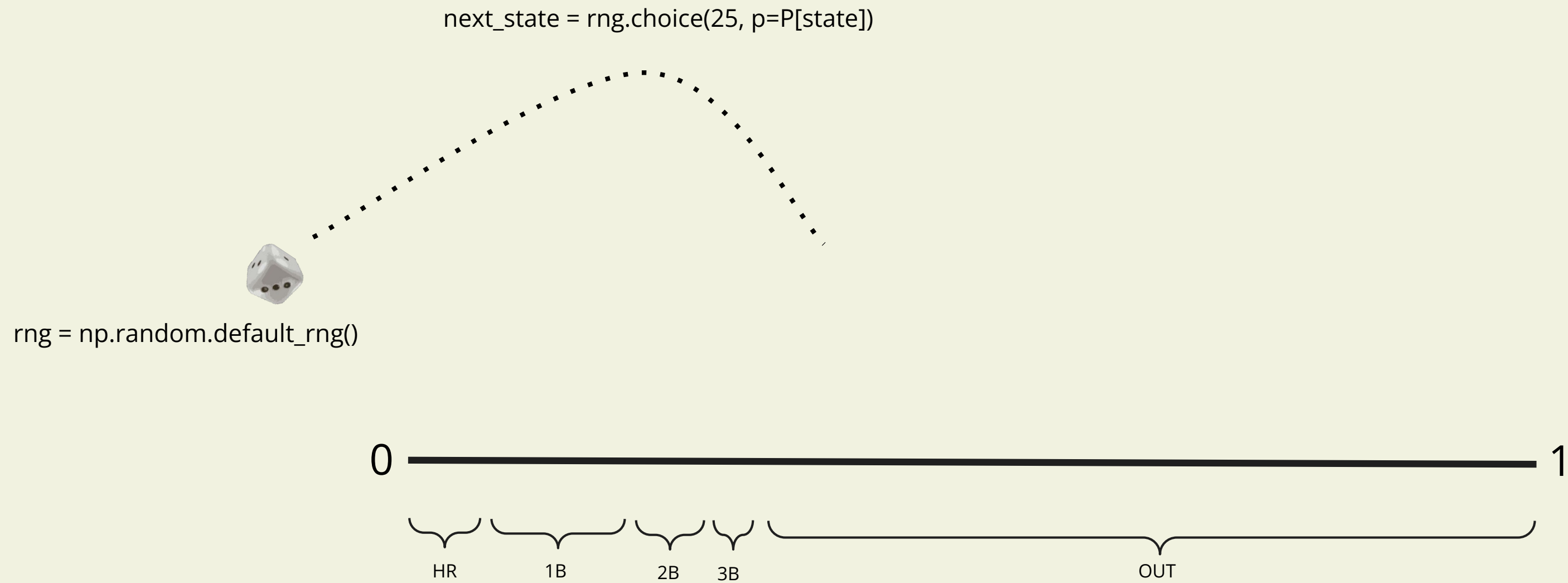
	0 outs, ---	0 outs, 1--	0 outs, -2-	0 outs, --3	0 outs, 12-	0 outs, 1-3	0 outs, -23	0 outs,123	1 out, ---	1 out, 1--	1 out, -2-	1 out, --3	1 out, 12-	1 out, 1-3	1 out, -23	1 out, 123	2 outs, ---	2 outs, 1--	2 outs, -2-	2 outs, --3	2 outs, 12-	2 outs, 1-3	2 outs, -23	2 outs,123	3 outs (inning over)	
0 outs, ---	0.0362	0.235	0.043	0.005	0	0	0	0	0.6808	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0 outs, 1--	0.0362	0	0.026	0.005	0.235	0	0.043	0	0	0.64312	0	0	0	0	0	0	0.01168	0	0	0	0	0	0	0	0	1
0 outs, -2-	0.0362	0	0.0433	0.005	0.097	0.1383	0	0	0	0	0.6808	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1.0006
0 outs, --3	0.0362	0.1383	0.0433	0.005	0	0.097	0	0	0	0	0	0.6808	0	0	0	0	0	0	0	0	0	0	0	0	0	1.0006
0 outs, 12-	0.0362	0	0	0.005	0	0	0.0433	0.235	0	0	0	0	0.66912	0	0	0	0	0.01168	0	0	0	0	0	0	0	1.0003
0 outs, 1-3	0.0362	0	0	0.005	0.1382	0	0.0693	0.097	0	0	0	0	0	0.64312	0	0	0	0	0	0.01168	0	0	0	0	0	1.0005
0 outs, -23	0.0362	0	0.0433	0.005	0	0.1382	0	0.097	0	0	0	0	0	0	0.6808	0	0	0	0	0	0	0	0	0	0	1.0005
0 outs, 123	0.0362	0	0	0.005	0	0	0.0433	0.235	0	0	0	0	0	0	0	0.6808	0	0	0	0	0	0	0	0	0	1.0003
1 out, ---	0	0	0	0	0	0	0	0	0.0362	0.235	0.0433	0.005	0	0	0	0	0.6808	0	0	0	0	0	0	0	0	1.0003
1 out, 1--	0	0	0	0	0	0	0	0	0.0362	0	0.026	0.005	0.235	0	0.0433	0	0	0.64312	0	0	0	0	0	0	0.01168	1.0003
1 out, -2-	0	0	0	0	0	0	0	0	0.0362	0	0.0433	0.005	0.097	0.1383	0	0	0	0	0.6808	0	0	0	0	0	0	1.0006
1 out, --3	0	0	0	0	0	0	0	0	0.0362	0.1383	0.0433	0.005	0	0.097	0	0	0	0	0	0.6808	0	0	0	0	0	1.0006
1 out, 12-	0	0	0	0	0	0	0	0	0.0362	0	0	0.005	0	0	0.0433	0.235	0	0	0	0	0.66912	0	0	0	0.01168	1.0003
1 out, 1-3	0	0	0	0	0	0	0	0	0.0362	0	0	0.005	0.1383	0	0.0693	0.097	0	0	0	0	0	0.64312	0	0	0.01168	1.0006
1 out, -23	0	0	0	0	0	0	0	0	0.0362	0	0.0433	0.005	0	0.1383	0	0.097	0	0	0	0	0	0	0.6808	0	0	1.0006
1 out, 123	0	0	0	0	0	0	0	0	0.0362	0	0	0.005	0	0	0.0433	0.235	0	0	0	0	0	0	0	0.66912	0.01168	1.0003
2 outs, ---	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0.235	0.0433	0.005	0	0	0	0	0.6808	1.0003
2 outs, 1--	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0	0.026	0.005	0.235	0	0.0433	0	0.6548	1.0003
2 outs, -2-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0	0.0433	0.005	0.097	0.1383	0	0	0.6808	1.0006
2 outs, --3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0.1383	0.0433	0.005	0	0.097	0	0	0.6808	1.0006
2 outs, 12-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0	0	0.005	0	0	0.0433	0.235	0.6808	1.0003
2 outs, 1-3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0	0	0.005	0.1383	0	0.0693	0.097	0.6548	1.0006
2 outs, -23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0	0.0433	0.005	0	0.1383	0	0.097	0.6808	1.0006
2 outs, 123	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0362	0	0	0.005	0	0	0.0433	0.235	0.6808	1.0003
3 outs (inning over)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

	0 outs, ---	0 outs, 1--	0 outs, -2-	0 outs, --3	0 outs, 12-	0 outs, 1-3	0 outs, -23	0 outs, 123	1 out, ---	1 out, 1--	1 out, -2-	1 out, --3	1 out, 12-	1 out, 1-3	1 out, -23	1 out, 123	2 outs, ---	2 outs, 1--	2 outs, -2-	2 outs, --3	2 outs, 12-	2 outs, 1-3	2 outs, -23	2 outs, 123	3 outs (inning over)
0 outs, ---	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 outs, 1--	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 outs, -2-	2	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 outs, --3	2	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 outs, 12-	3	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 outs, 1-3	3	0	0	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 outs, -23	3	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 outs, 123	4	0	0	3	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1 out, ---	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1 out, 1--	0	0	0	0	0	0	0	0	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1 out, -2-	0	0	0	0	0	0	0	0	2	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1 out, --3	0	0	0	0	0	0	0	0	2	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1 out, 12-	0	0	0	0	0	0	0	0	3	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0
1 out, 1-3	0	0	0	0	0	0	0	0	3	0	0	0	2	0	0	1	0	0	0	0	0	0	0	0	0
1 out, -23	0	0	0	0	0	0	0	0	3	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0
1 out, 123	0	0	0	0	0	0	0	0	4	0	0	0	3	0	0	0	1	0	0	0	0	0	0	0	0
2 outs, ---	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
2 outs, 1--	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	0	0	0	0
2 outs, -2-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	1	1	0	0	0	0
2 outs, --3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	1	1	0	0	0	0
2 outs, 12-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	2	0	0	0	0
2 outs, 1-3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	2	0	0	1	0
2 outs, -23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	2	0	0	0	0
2 outs, 123	0	0																							

MONTE CARLO SIMULATION



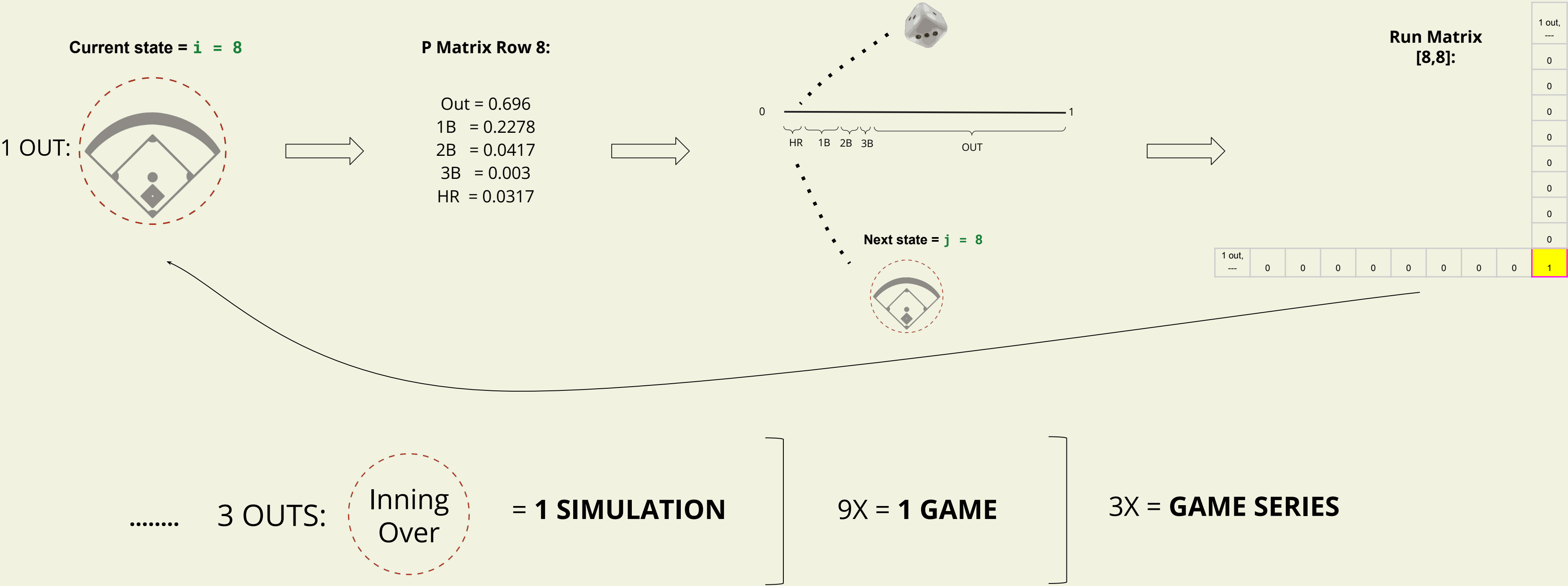
BASIC IDEA:



Monte Carlo Simulation



EXAMPLE (ORIOLES):



RESULTS 1



GAME 1

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	1.0	1.0	0.0
1B	Orioles	0.0	1.0	0.0
2T	Cubs	0.0	1.0	0.0
2B	Orioles	0.0	1.0	0.0
3T	Cubs	1.0	2.0	0.0
3B	Orioles	0.0	2.0	0.0
4T	Cubs	0.0	2.0	0.0
4B	Orioles	2.0	2.0	2.0
5T	Cubs	3.0	5.0	2.0
5B	Orioles	0.0	5.0	2.0
6T	Cubs	0.0	5.0	2.0
6B	Orioles	1.0	5.0	3.0
7T	Cubs	0.0	5.0	3.0
7B	Orioles	0.0	5.0	3.0
8T	Cubs	0.0	5.0	3.0
8B	Orioles	0.0	5.0	3.0
9T	Cubs	0.0	5.0	3.0
9B	Orioles	0.0	5.0	3.0

Game 1 Final: **Cubs 5.0** - Orioles 3.0

GAME 2

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	2.0	2.0	0.0
1B	Orioles	0.0	2.0	0.0
2T	Cubs	0.0	2.0	0.0
2B	Orioles	1.0	2.0	1.0
3T	Cubs	1.0	3.0	1.0
3B	Orioles	0.0	3.0	1.0
4T	Cubs	0.0	3.0	1.0
4B	Orioles	0.0	3.0	1.0
5T	Cubs	0.0	3.0	1.0
5B	Orioles	0.0	3.0	1.0
6T	Cubs	0.0	3.0	1.0
6B	Orioles	0.0	3.0	1.0
7T	Cubs	3.0	6.0	1.0
7B	Orioles	0.0	6.0	1.0
8T	Cubs	0.0	6.0	1.0
8B	Orioles	3.0	6.0	4.0
9T	Cubs	0.0	6.0	4.0
9B	Orioles	3.0	6.0	7.0

Game 2 Final: Cubs 6.0 - **Orioles 7.0**

GAME 3

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	0.0	0.0	0.0
1B	Orioles	1.0	0.0	1.0
2T	Cubs	0.0	0.0	1.0
2B	Orioles	0.0	0.0	1.0
3T	Cubs	0.0	0.0	1.0
3B	Orioles	0.0	0.0	1.0
4T	Cubs	1.0	1.0	1.0
4B	Orioles	4.0	1.0	5.0
5T	Cubs	0.0	1.0	5.0
5B	Orioles	0.0	1.0	5.0
6T	Cubs	0.0	1.0	5.0
6B	Orioles	0.0	1.0	5.0
7T	Cubs	4.0	5.0	5.0
7B	Orioles	0.0	5.0	5.0
8T	Cubs	4.0	9.0	5.0
8B	Orioles	0.0	9.0	5.0
9T	Cubs	0.0	9.0	5.0
9B	Orioles	0.0	9.0	5.0

Game 3 Final: **Cubs 9.0** - Orioles 5.0

SERIES RESULTS

Cubs wins: 2
Orioles wins: 1

SERIES WINNER: Cubs (2-1)

RESULTS 2



GAME 1

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	0.0	0.0	0.0
1B	Orioles	2.0	0.0	2.0
2T	Cubs	0.0	0.0	2.0
2B	Orioles	0.0	0.0	2.0
3T	Cubs	2.0	2.0	2.0
3B	Orioles	0.0	2.0	2.0
4T	Cubs	0.0	2.0	2.0
4B	Orioles	0.0	2.0	2.0
5T	Cubs	0.0	2.0	2.0
5B	Orioles	2.0	2.0	4.0
6T	Cubs	0.0	2.0	4.0
6B	Orioles	0.0	2.0	4.0
7T	Cubs	0.0	2.0	4.0
7B	Orioles	0.0	2.0	4.0
8T	Cubs	0.0	2.0	4.0
8B	Orioles	1.0	2.0	5.0
9T	Cubs	2.0	4.0	5.0
9B	Orioles	0.0	4.0	5.0

Game 1 Final: Cubs 4.0 - Orioles 5.0

GAME 2

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	1.0	1.0	0.0
1B	Orioles	3.0	1.0	3.0
2T	Cubs	0.0	1.0	3.0
2B	Orioles	0.0	1.0	3.0
3T	Cubs	0.0	1.0	3.0
3B	Orioles	1.0	1.0	4.0
4T	Cubs	0.0	1.0	4.0
4B	Orioles	0.0	1.0	4.0
5T	Cubs	0.0	1.0	4.0
5B	Orioles	1.0	1.0	5.0
6T	Cubs	0.0	1.0	5.0
6B	Orioles	0.0	1.0	5.0
7T	Cubs	4.0	5.0	5.0
7B	Orioles	0.0	5.0	5.0
8T	Cubs	2.0	7.0	5.0
8B	Orioles	0.0	7.0	5.0
9T	Cubs	0.0	7.0	5.0
9B	Orioles	0.0	7.0	5.0

Game 2 Final: Cubs 7.0 - Orioles 5.0

Game 3 (OT)

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	0.0	0.0	0.0
1B	Orioles	0.0	0.0	0.0
2T	Cubs	0.0	0.0	0.0
2B	Orioles	0.0	0.0	0.0
3T	Cubs	1.0	1.0	0.0
3B	Orioles	0.0	1.0	0.0
4T	Cubs	0.0	1.0	0.0
4B	Orioles	0.0	1.0	0.0
5T	Cubs	1.0	2.0	0.0
5B	Orioles	0.0	2.0	0.0
6T	Cubs	0.0	2.0	0.0
6B	Orioles	0.0	2.0	0.0
7T	Cubs	0.0	2.0	0.0
7B	Orioles	2.0	2.0	2.0
8T	Cubs	0.0	2.0	2.0
8B	Orioles	0.0	2.0	2.0
9T	Cubs	0.0	2.0	2.0
9B	Orioles	0.0	2.0	2.0
10T	Cubs	0.0	2.0	2.0
10B	Orioles	1.0	2.0	3.0

Game 3 Final: Cubs 2.0 - Orioles 3.0

SERIES RESULTS

Cubs wins: 1
Orioles wins: 2

SERIES WINNER: Orioles (2-1)

RESULTS 3



GAME 1

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	1.0	1.0	0.0
1B	Orioles	0.0	1.0	0.0
2T	Cubs	2.0	3.0	0.0
2B	Orioles	0.0	3.0	0.0
3T	Cubs	1.0	4.0	0.0
3B	Orioles	0.0	4.0	0.0
4T	Cubs	0.0	4.0	0.0
4B	Orioles	0.0	4.0	0.0
5T	Cubs	0.0	4.0	0.0
5B	Orioles	0.0	4.0	0.0
6T	Cubs	0.0	4.0	0.0
6B	Orioles	1.0	4.0	1.0
7T	Cubs	0.0	4.0	1.0
7B	Orioles	0.0	4.0	1.0
8T	Cubs	0.0	4.0	1.0
8B	Orioles	0.0	4.0	1.0
9T	Cubs	2.0	6.0	1.0
9B	Orioles	0.0	6.0	1.0

Game 1 Final: **Cubs 6.0** - Orioles 1.0

GAME 2

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	0.0	0.0	0.0
1B	Orioles	0.0	0.0	0.0
2T	Cubs	0.0	0.0	0.0
2B	Orioles	1.0	0.0	1.0
3T	Cubs	5.0	5.0	1.0
3B	Orioles	0.0	5.0	1.0
4T	Cubs	0.0	5.0	1.0
4B	Orioles	0.0	5.0	1.0
5T	Cubs	0.0	5.0	1.0
5B	Orioles	0.0	5.0	1.0
6T	Cubs	0.0	5.0	1.0
6B	Orioles	0.0	5.0	1.0
7T	Cubs	0.0	5.0	1.0
7B	Orioles	1.0	5.0	2.0
8T	Cubs	0.0	5.0	2.0
8B	Orioles	0.0	5.0	2.0
9T	Cubs	0.0	5.0	2.0
9B	Orioles	0.0	5.0	2.0

Game 2 Final: **Cubs 5.0** - Orioles 2.0

GAME 3

Inning	Team	Runs_Scored	Cubs_Total	Orioles_Total
1T	Cubs	2.0	2.0	0.0
1B	Orioles	0.0	2.0	0.0
2T	Cubs	1.0	3.0	0.0
2B	Orioles	0.0	3.0	0.0
3T	Cubs	1.0	4.0	0.0
3B	Orioles	0.0	4.0	0.0
4T	Cubs	1.0	5.0	0.0
4B	Orioles	0.0	5.0	0.0
5T	Cubs	4.0	9.0	0.0
5B	Orioles	1.0	9.0	1.0
6T	Cubs	0.0	9.0	1.0
6B	Orioles	1.0	9.0	2.0
7T	Cubs	0.0	9.0	2.0
7B	Orioles	2.0	9.0	4.0
8T	Cubs	0.0	9.0	4.0
8B	Orioles	1.0	9.0	5.0
9T	Cubs	0.0	9.0	5.0
9B	Orioles	0.0	9.0	5.0

Game 3 Final: **Cubs 9.0** - Orioles 5.0

SERIES RESULTS

Cubs wins: 3
Orioles wins: 0

SERIES WINNER: Cubs (3-0)

10,000 SIMULATIONS



- ⚾ **Average Cubs runs per game: 3.89**
- ⚾ **Average Orioles runs per game: 3.31**
- ⚾ **Cubs win percentage: 50.44%**
- ⚾ **Orioles win percentage: 38.60%**



INTERPRETATION



- 🏈 Cubs are expected to win in a three-game series with a slight advantage
 - Ran 10 series simulations: Cubs won 6 of them
 - Ran 10,000 game simulations: Cubs had higher runs per game & win percentage
- 🏈 Consistent with what we had expected prior to our modeling process, due to several factors:
 - Cubs record (92-70) vs. Orioles record (75-87)
 - Cubs' higher probabilities of walks, singles, doubles, triples, hits, stolen bases, and home runs
 - Cubs' lower probabilities of strikeouts, grounded into double plays, and outs

ASSESSMENT



Overall: Model performed well in giving results that reflect 2025 batting statistics

Limitations:

- 🏈 Conservative baserunning (could underestimate runs or outs)
- 🏈 Slightly Inflated Runs → steals successful all the time → could include caught stealing
- 🏈 Doesn't take into consideration real-game situations (home-away, injuries/fatigue, nerves in clutch situations) → could affect player performance, and therefore the game's outcome
- 🏈 Doesn't consider the new rule of "ghost runners" in extra innings

CITATIONS



-  "2025 Baltimore Orioles Statistics." *Baseball Reference*,
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THANK YOU!

QUESTIONS?

